



COURSE SLIDES

Agentic AI for Digital Marketing and Advertising (C1798)

Course Code: C1798

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

About the Trainer



Courseware Team

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Industry certified in the subject area of this course.

DELIVERS

Professional short courses for Tertiary Infotech Academy.

ORGANISATION

Tertiary Infotech Academy Pte Ltd

Let's Know Each Other

1 Your name and organisation / role.

2 Your experience with this subject (if any).

3 What you want to be able to do after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

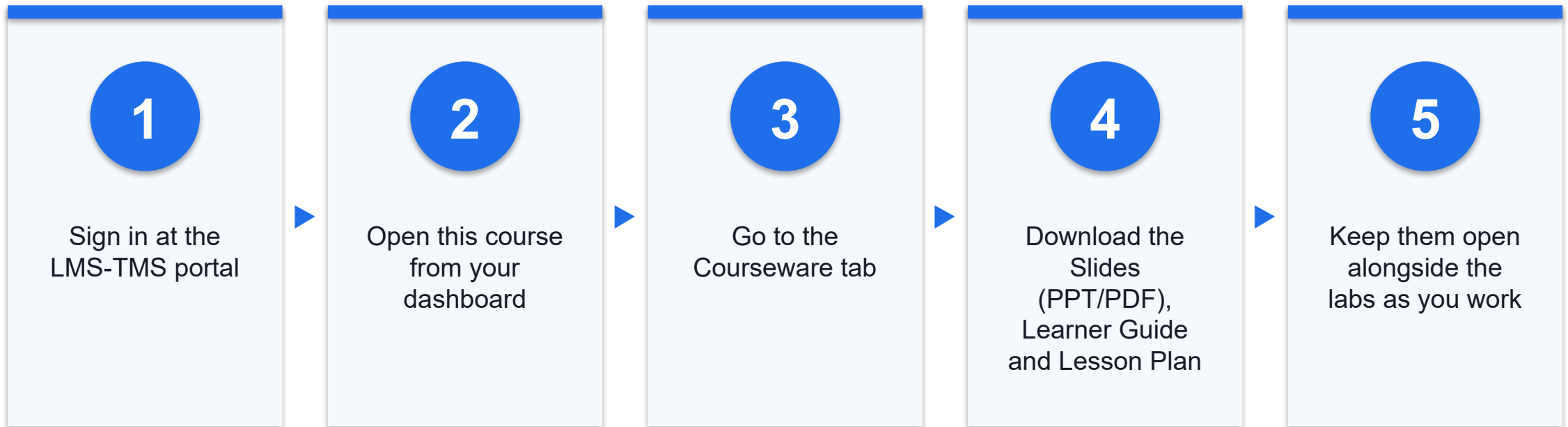
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Download Course Material



Portal: <https://lms-tms.tertiaryinfotech.com>

Lesson Plan — 2 Days, 7.5 hours/day

Day 1

- Day 1 — Agent Foundations, Trend Intelligence, and Content Automation
 - Topic 1: Foundations of Agentic AI for Marketing (Labs 1–2)
 - Topic 2: Automated Trend Discovery and Viral Content Research (Labs 3–4)
 - Topic 3: AI Content Creation and Multimedia Automation (Labs 5–6)

Day 2

- Day 2 — Campaign Orchestration, Performance Learning, and Controlled Optimization
 - Topic 4: Creating and Launching AI-Driven Marketing Campaigns (Labs 7–8)
 - Topic 5: Agentic AI for Social Media Performance Monitoring (Labs 9–10)
 - Topic 6: Agentic AI for Advertising Optimization and Scaling (Labs 11–12)
- Daily timing
 - 9:30 am - 5:30 pm (30-minute lunch; tea breaks within scheduled time)

Learning Outcomes

1

Agent Architecture

Explain agentic marketing architecture and define goals, tools, memory, approval gates, and operational guardrails for a marketing agent.

2

Trend Intelligence

Build a repeatable trend-discovery workflow that evaluates freshness, evidence, brand fit, and risk before recommending an opportunity.

3

Content Automation

Produce an aligned multi-platform content and multimedia kit that preserves audience intent, brand voice, factual accuracy, and creative rights.

4

Campaign Orchestration

Design an AI-assisted campaign blueprint, experiment plan, content calendar, and controlled launch workflow tied to objectives and KPIs.

5

Performance Learning

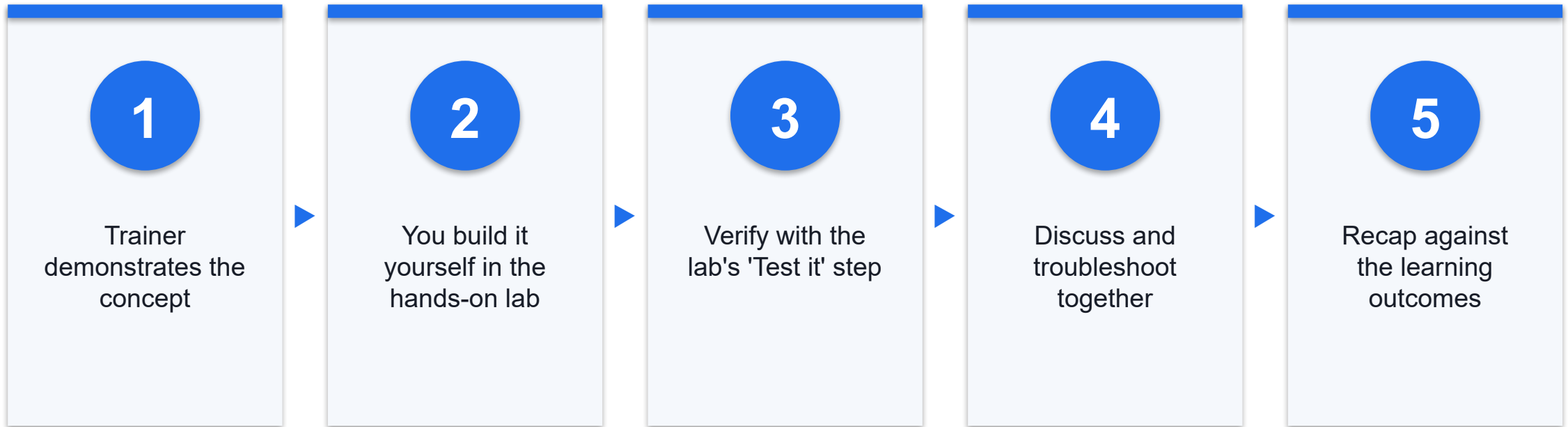
Calculate and interpret social performance metrics, analyse synthetic audience feedback, and turn findings into evidence-linked recommendations.

6

Optimization Control

Diagnose paid-media performance, propose bounded experiments, and design a human-governed optimization loop with stop conditions and audit evidence.

How You'll Learn





CORE CONCEPTS

Agentic Marketing Fundamentals

Six Ideas That Anchor the Course

1

Agent

A goal-directed system that can choose among permitted actions and tools.

2

Workflow

A defined path for reliable, repeatable transformations and checks.

3

State

The versioned campaign facts, decisions, and results carried across steps.

4

Tool

A bounded capability to read, calculate, draft, schedule, or act.

5

Guardrail

A testable rule that blocks, routes, limits, or records an action.

6

Feedback

Evidence used to update the next decision rather than repeat the same plan.

The GSRAL Control Loop

1

Goal

Define business outcome, audience, KPI, constraints, and owner.

2

Sense

Collect approved signals with source, time, market, and quality.

3

Reason

Compare options against evidence, rules, risk, and uncertainty.

4

Act

Draft or execute only the action permitted for the current state.

5

Learn

Record result, exception, feedback, and the next-cycle state.

6

Approval gate

Human authorization before public, personal-data, targeting, or spend actions.

CORE IDEA

Autonomy is a permission design, not a personality trait.

Give the system freedom where errors are cheap and reversible; require human judgement where claims, privacy, reputation, or money are at stake.

The Connected Campaign You Will Build

1

Intelligence

Agent canvas and guardrails · Trend signal table ·
Ranked opportunity brief

2

Creative

Platform content kit · Image concept · Short-video
storyboard

3

Launch

Campaign and KPI blueprint · Experiment map ·
Controlled content calendar

4

Learning

Performance dashboard · Feedback themes ·
Optimization runbook

How Every Lab Progresses

1

Start from the current Aurora Active campaign checkpoint.

2

Apply the topic model to a bounded marketing task.

3

Create a versioned artifact with sources and assumptions.

4

Verify the artifact against an observable checklist or formula.

5

Carry the approved checkpoint into the next lab.

TOPIC 01

Foundations of Agentic AI for Marketing

Agents and workflows | goal design | tools and memory | approval gates | brand, privacy, and advertising guardrails

KEY CONCEPTS

Key Concepts — Foundations of Agentic AI for Marketing

1

An agent combines a goal, state, reasoning, tools, and feedback; a fixed workflow follows a predefined path.

2

A marketing pipeline should make each hand-off explicit: inputs, decision rule, output schema, owner, and failure route.

3

Autonomy is not all-or-nothing. Read, draft, recommend, approve, publish, and spend actions carry different risk.

4

Memory must be scoped: campaign facts and approved preferences belong in state; sensitive or unverified data does not.

5

Tool permissions should follow least privilege, with credentials stored in a secure connection rather than a prompt or repository.

6

A useful guardrail is observable: it defines what is blocked, when review is required, and what evidence is logged.

The Agentic Marketing Control Loop

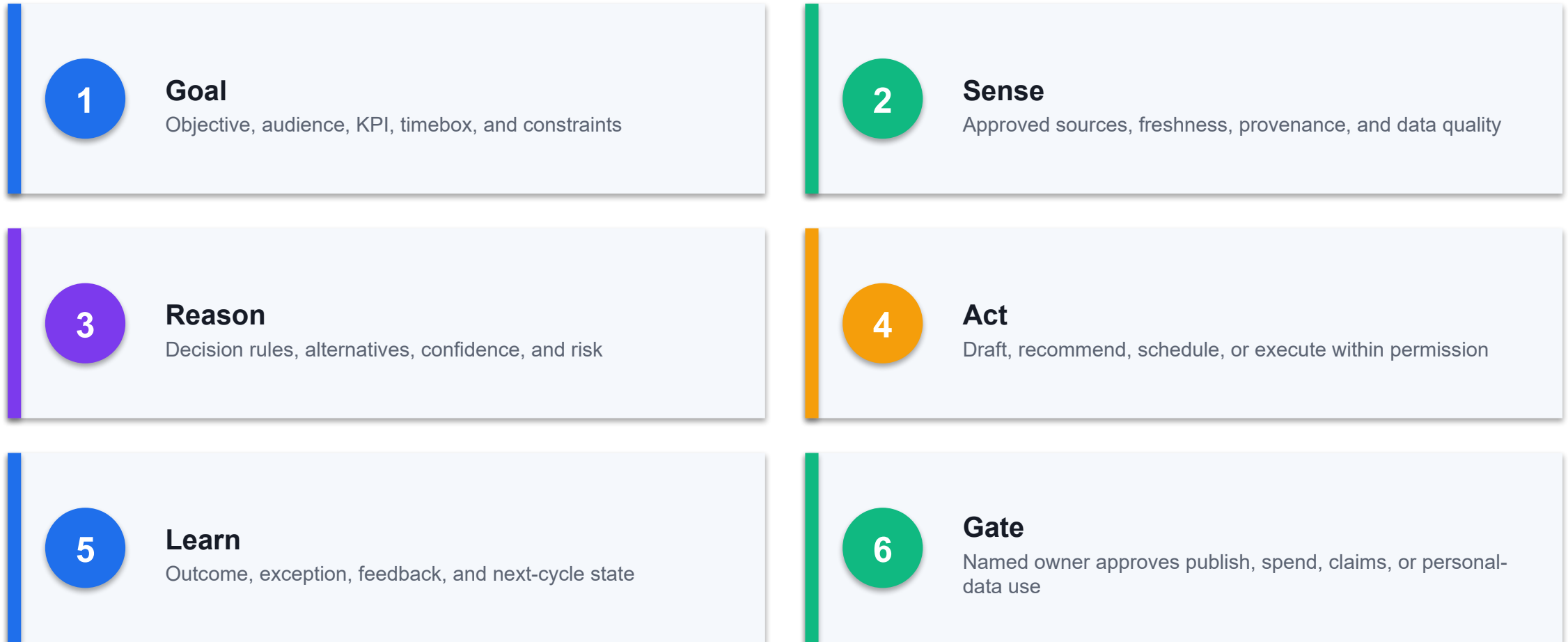
1

A reliable agent moves through five observable states: Goal, Sense, Reason, Act, and Learn. Goal defines the outcome and constraints. Sense collects permitted signals. Reason compares options against rules. Act produces only an allowed output. Learn records the result so the next cycle starts with better evidence.

2

Approval gates sit between reasoning and high-impact actions. They are not delays added after the design; they are part of the control loop. The agent should also have a stop path for missing data, conflicting instructions, or a threshold breach.

Visual Model — The Agentic Marketing Control Loop



Workflow or Agent? Choose by Uncertainty

1

Use a fixed workflow when the path is predictable and the cost of variation is high. Examples include formatting a weekly report or moving approved copy into a calendar. Use agentic reasoning when inputs vary and the system must compare several valid routes, such as ranking trend signals with incomplete evidence.

2

Many production systems are hybrids: deterministic collection and calculation feed an agent that explains patterns, then a deterministic approval and logging step controls the outcome. The design should be no more autonomous than the task requires.

Visual Model — Workflow or Agent? Choose by Uncertainty

1

Fixed workflow

Known steps, stable inputs, repeatable output

2

Router

Chooses one approved path from explicit categories

3

Evaluator loop

Generates, checks, and revises against a rubric

4

Agent

Plans among tools when the path cannot be fixed in advance

Guardrails as Testable Controls

1

A policy statement such as 'be accurate' is too vague to operate. A control translates policy into a condition and response: unsupported product claims are blocked, synthetic audience data is required in training, and a named owner must approve any public or paid action.

2

Controls need evidence. Log the input source, decision, output version, approver, timestamp, and reason. This record supports troubleshooting and makes it possible to decide whether the system should continue, pause, or be changed.

Visual Model — Guardrails as Testable Controls

1

Claim check

Block any claim without a source and owner

2

Privacy check

Reject identifiable customer data in prompts

3

Brand check

Compare tone, exclusions, and visual direction

4

Action check

Require approval for publish, spend, or targeting changes

Hands-On Labs — Foundations of Agentic AI for Marketing

Lab 1

- Map the Aurora Agentic Marketing Control Loop

Lab 2

- Write and Test the Marketing Agent Operating Specification

Map the Aurora Agentic Marketing Control Loop

LAB 1

You turn the Aurora Active launch brief into a visible Goal-Sense-Reason-Act-Learn control loop. You decide which actions may run automatically, which remain draft-only, and which need a named human owner. The result becomes the architecture reference for all later labs.

You'll build

01-foundation/01-agent-canvas.md containing the campaign goal, state, inputs, decisions, outputs, permissions, approval gates, stop conditions, and audit fields.

Tools: Aurora brand brief, document editor, classroom-approved AI assistant

Build Path & Verification — Lab 1

1

1. Translate the campaign brief into Goal-Sense-Reason-Act-Learn states.

2

2. Assign read, calculate, draft, recommend, approve, or disabled permissions.

3

3. Add observable stop conditions and a complete audit record.

4

Verify: Open 01-agent-canvas.md and point to all five loop stages, at least six permissioned actions, three approval-gated actions, six stop conditions with safe responses, and the ten audit fields. EXECUTE must be disabled and the campaign goal must match the Aurora brief.

Write and Test the Marketing Agent Operating Specification

LAB 2

You convert the control canvas into an operating specification that an AI assistant can follow consistently. You define inputs, tools, memory, output schema, refusal and escalation rules, then run a normal case and an unsafe case to confirm the boundaries work.

You'll build

01-foundation/02-agent-operating-spec.md with a structured system instruction, tool policy, state schema, output schema, and two documented boundary tests.

Tools: Aurora brand brief, Lab 1 agent canvas, document editor, classroom-approved AI assistant

Build Path & Verification — Lab 2

1

1. Define approved inputs, fictional tools, campaign state, and output schema.

2

2. Run one normal request and one deliberately unsafe request.

3

3. Tighten the specification until the unsafe request routes to review.

4

Verify: Test A returns a structured DRAFT using only supplied facts. Test B returns NEEDS_HUMAN_REVIEW, rejects the unsupported health claim and personal-data request, and does not claim to publish or change spend. Both tests have versioned log entries.

Recap — Foundations of Agentic AI for Marketing

1

You can now: LO1: Explain agentic marketing architecture and define goals, tools, memory, approval gates, and operational guardrails for a marketing agent

TOPIC 02

Automated Trend Discovery and Viral Content Research

signal collection | source provenance | normalized trend data | opportunity scoring | hooks and emotional triggers | scheduled briefs

KEY CONCEPTS

Key Concepts — Automated Trend Discovery and Viral Content Research

1

A trend signal is an observation with a source, market, timestamp, topic, and evidence; a trend is not merely a popular-looking post.

2

Google Trends reports relative, normalized search interest rather than absolute search volume, so comparisons need matched regions and time windows.

3

Triangulation reduces false positives by comparing search, community, editorial, and first-party signals instead of trusting one source.

4

Viral mechanics include a clear hook, useful tension, social currency, emotional relevance, and a low-friction action, but none guarantees performance.

5

Opportunity scoring should separate upside from risk so a high-volume but unsafe idea cannot win by arithmetic alone.

6

A scheduled trend brief needs freshness limits, deduplication, citations, exception handling, and an explicit review queue.

From Raw Signals to a Trend Brief

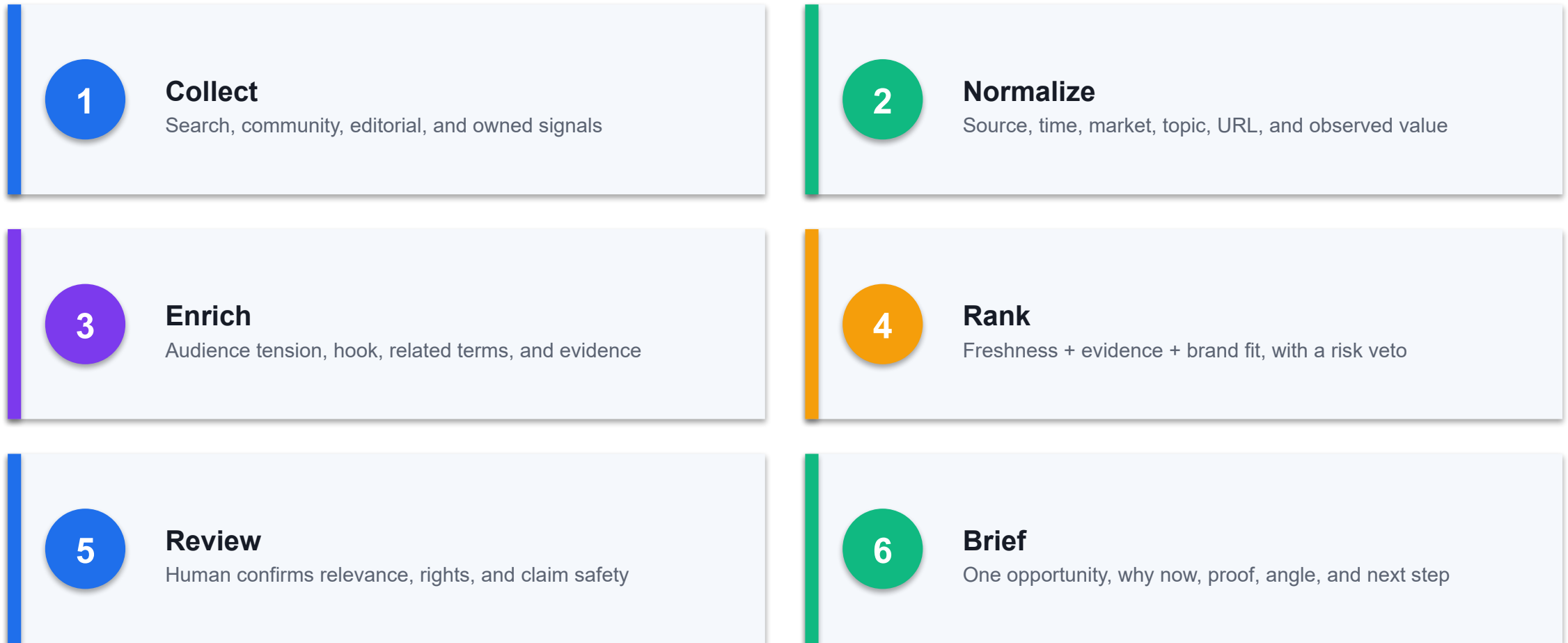
1

Collection gathers candidate signals from approved public or first-party sources. Normalization gives every record the same fields. Enrichment adds related queries, audience tension, hook, and evidence notes. Ranking compares opportunities. Review confirms that the recommendation is timely, truthful, and useful.

2

Each transformation should preserve the source URL and observation time. If an agent cannot cite the supporting record, the item should remain in a research queue rather than move to content production.

Visual Model — From Raw Signals to a Trend Brief



Read Relative Trend Data Correctly

1

A Google Trends score of 100 is the peak relative interest for the selected comparison, geography, and time range. It is not the number of searches. Changing the window or location changes the denominator, so screenshots from different settings should not be compared as if they shared a scale.

2

A spike is a prompt to investigate, not proof of demand or positive sentiment. Compare a candidate with a baseline, inspect related queries, confirm the timing in another source, and document uncertainty before recommending action.

Visual Model — Read Relative Trend Data Correctly

1

100

Peak share within the selected comparison

2

50

Half the relative interest of that peak, not half the searches

3

0

Insufficient data or very low relative interest, not necessarily none

4

Breakout

Very rapid growth that still needs context and evidence

Opportunity Score with a Risk Veto

1

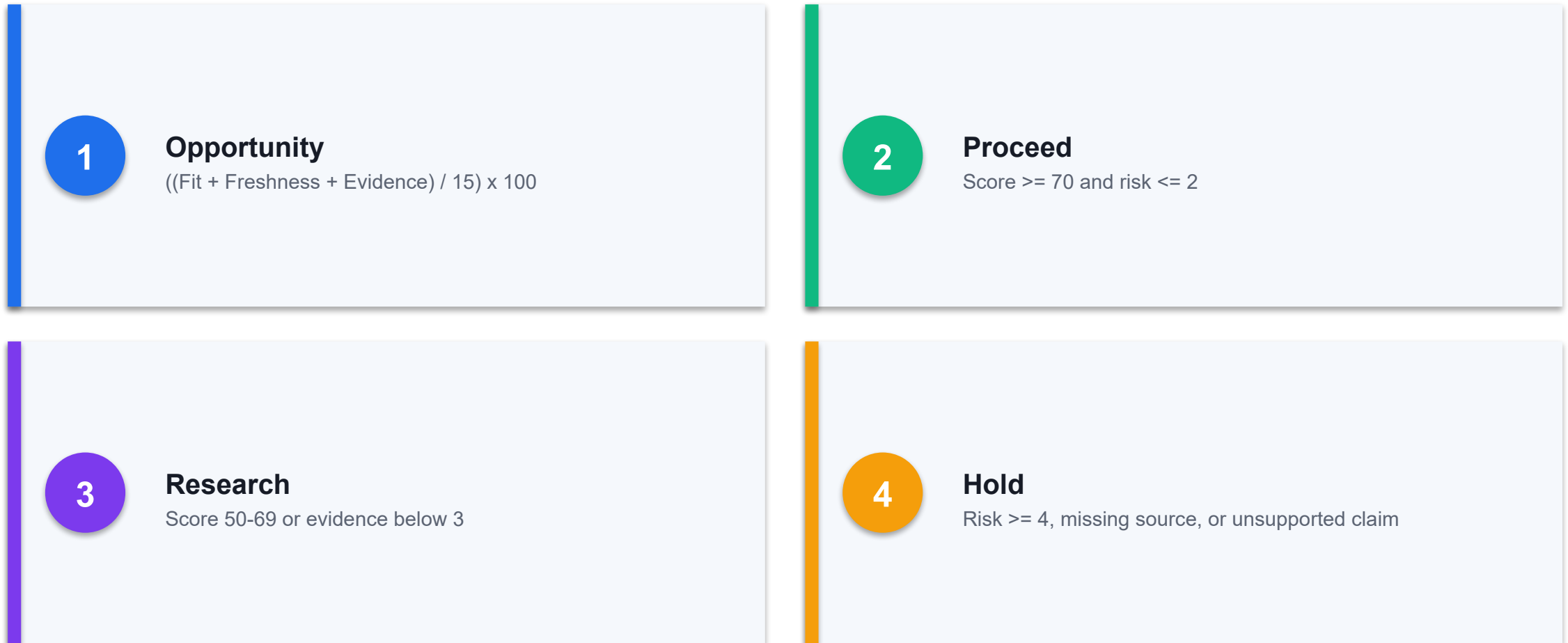
For classroom use, score brand fit, freshness, and evidence from 1 to 5. The opportunity score is their average multiplied by 20, producing a 0-100 scale. Keep risk separate. A risk score of 4 or 5 sends the signal to review regardless of the opportunity score.

2

Example: a leak-prevention conversation scores 5 for fit, 4 for freshness, and 3 for evidence. Its opportunity score is 80. With risk 2, it can move to a brief. A celebrity rumour can score high for freshness but risk 5, so the correct action is hold, verify, or reject.

WORKED EXAMPLE

Visual Model — Opportunity Score with a Risk Veto



Hands-On Labs — Automated Trend Discovery and Viral Content Research

Lab 3

- Build and Score a Source-Linked Trend Signal Table

Lab 4

- Generate a Viral Opportunity Brief and Scheduled Research Spec

Build and Score a Source-Linked Trend Signal Table

LAB 3

You normalize eight synthetic trend signals, calculate a transparent opportunity score, and apply a separate risk veto. You then inspect the strongest and riskiest records so the ranking is based on evidence rather than engagement alone.

You'll build

02-trends/03-trend-signal-scorecard.xlsx (or a native Google Sheet) with calculated opportunity_score and decision columns, plus a short data-quality note.

Tools: Spreadsheet application, Aurora trend signals CSV, Aurora brand brief

Build Path & Verification — Lab 3

1. Validate provenance and required fields for eight synthetic signals.

2. Calculate opportunity from fit, freshness, and evidence.

3. Apply an independent risk veto and document data limitations.

4. Verify: The workbook contains exactly eight signals. T01 has opportunity_score 93 and decision PROCEED. T07 has opportunity_score 47 and decision HOLD. No row with risk 4 or 5 is labelled PROCEED, and the Data Quality sheet states the formula and limitations.

Generate a Viral Opportunity Brief and Scheduled Research Spec

LAB 4

You use only the ranked records to create an evidence-linked trend opportunity brief. You then specify a weekly research workflow with a fixed schedule, required fields, deduplication, expiry, and exception route so the process can be automated safely later.

You'll build

02-trends/04-viral-opportunity-brief.md and 02-trends/04-weekly-trend-workflow-spec.md, both linked to the Lab 3 signal IDs and review rules.

Tools: Ranked trend CSV, Aurora brief, Lab 2 operating specification, classroom-approved AI assistant

Build Path & Verification — Lab 4

1

1. Turn three ranked signals into one evidence-linked opportunity.

2

2. Review every claim and mark missing product evidence explicitly.

3

3. Specify schedule, freshness, deduplication, exceptions, and versioned output.

4

Verify: The opportunity brief recommends one angle, cites only T01/T03/T06, separates observation and interpretation, and marks the product demonstration for review. The workflow spec says Monday 08:30 Asia/Singapore, 7-day lookback, 72-hour active limit, a deduplication key, six exception routes, and a versioned output schema.

Recap — Automated Trend Discovery and Viral Content Research

1

You can now: LO2: Build a repeatable trend-discovery workflow that evaluates freshness, evidence, brand fit, and risk before recommending an opportunity

TOPIC 03

AI Content Creation and Multimedia Automation

message hierarchy | platform adaptation | brand-voice prompting | image generation | short-form video | scheduling and publish gates

Key Concepts — AI Content Creation and Multimedia Automation

1

A message hierarchy keeps one campaign promise intact while allowing the hook, proof, format, and call to action to change by platform.

2

Platform adaptation is not copy-and-paste resizing; audience intent, reading behavior, and native format change the expression of the idea.

3

A reusable prompt separates immutable brand facts from variable audience, channel, format, and experiment fields.

4

Image prompts need subject, setting, composition, lighting, palette, exclusions, and output format; generated text inside images still needs checking.

5

A short-form video pipeline connects hook, scene, on-screen text, voiceover, caption, safe-zone, and rights checks.

6

Automation should create and route drafts; publishing stays gated until claims, brand fit, accessibility, and platform readiness are verified.

One Message, Four Platform Expressions

1

The campaign promise should remain recognisable across Facebook, Instagram, LinkedIn, and X. What changes is the opening, depth, visual rhythm, and call to action. Instagram may lead with a visual payoff, LinkedIn with a workplace use case, Facebook with context and conversation, and X with one sharp observation.

2

The agent should output a structured content object rather than an unlabelled paragraph. Fields such as platform, audience, hook, proof, caption, CTA, accessibility note, and claim source make review and scheduling safer.

Visual Model — One Message, Four Platform Expressions

1

Instagram

Visual-first hook, concise caption, save/share value

2

Facebook

Context, practical benefit, and conversation prompt

3

LinkedIn

Professional problem, evidence, and operational takeaway

4

X

One idea, timely framing, and a direct link or reply prompt

Brand Prompt Anatomy

1

A robust content prompt has five layers: verified facts, audience and job-to-be-done, brand voice, output schema, and review constraints. Keeping these layers visible makes the prompt easier to update and audit than a long paragraph of mixed instructions.

2

Ask the model to flag missing evidence instead of filling gaps. Require a claim ledger that lists each factual statement and its source. This turns hallucination checking into a normal production step rather than an afterthought.

Visual Model — Brand Prompt Anatomy

1

Facts

Only approved product, offer, and timing details

2

Audience

Segment, situation, tension, and desired action

3

Voice

Tone, sentence style, vocabulary, and exclusions

4

Schema

Named fields for each platform and asset

5

Checks

Claims, accessibility, rights, and approval status

6

Uncertainty

Write NEEDS EVIDENCE instead of inventing support

Multimedia Production with Review Gates

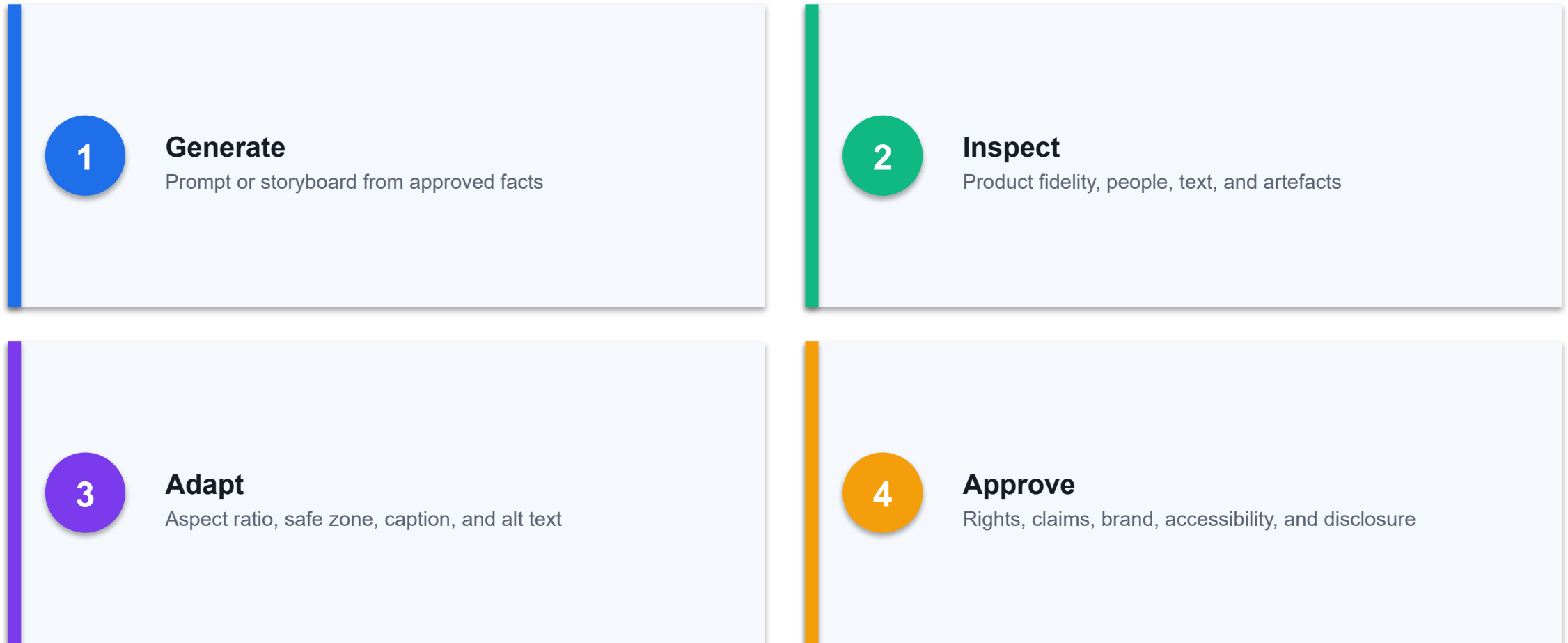
1

An image-generation request should describe the subject, setting, composition, lighting, colour palette, camera view, and exclusions. For product work, use a real approved product reference only when rights and platform terms permit it; otherwise create a clearly labelled concept image.

2

A short video can be planned as six connected fields per scene: time, visual, action, voiceover, on-screen text, and sound. Before scheduling, check visual continuity, text accuracy, captions, safe zones, rights, and whether any synthetic media disclosure is required by policy or platform.

Visual Model — Multimedia Production with Review Gates



Hands-On Labs — AI Content Creation and Multimedia Automation

Lab 5

- Create a Governed Four-Platform Content Kit

Lab 6

- Design a Branded Image Concept and Short-Form Video Storyboard

Create a Governed Four-Platform Content Kit

LAB 5

You turn one selected, review-gated trend angle into Facebook, Instagram, LinkedIn, and X drafts without changing the core product truth. A claim ledger and per-platform review fields make every variant traceable and ready for human review.

You'll build

03-content/05-four-platform-content-kit.md containing a message hierarchy, four platform drafts, accessibility notes, a claim ledger, and a final review status.

Tools: Aurora brief, Lab 4 opportunity brief, document editor, classroom-approved AI assistant

Build Path & Verification — Lab 5

1

1. Lock the audience, promise, proof, offer, and CTA in a message hierarchy.

2

2. Generate four native platform expressions with a fixed output schema.

3

3. Audit claims, accessibility, and cross-channel consistency before review.

4

Verify: Four drafts contain all nine fields, link claims to the ledger, use accessible alt text, and stay DRAFT. v0.1 preserves the complete baseline; v0.2 contains accepted evidence-backed edits; the Review Log records accepted and rejected decisions.

Design a Branded Image Concept and Short-Form Video Storyboard

LAB 6

You translate the campaign message into a reviewable image concept and a 15-second vertical video storyboard. You practise prompt anatomy, visual continuity, captions, safe zones, and rights review while keeping the product image conceptual and the output unpublished.

You'll build

03-content/06-creative-board.md plus an optional 06-hydraloop-concept.png, containing the image prompt, generation record, review checklist, six-scene storyboard, caption, and alt text.

Tools: Gemini image generation or Canva, document editor, Lab 5 content kit

Build Path & Verification — Lab 6

1

1. Translate the approved message into a bounded image prompt or wireframe.

2

2. Review product fidelity, claims, rights, accessibility, and artefacts.

3

3. Build a six-scene, 15-second storyboard with captions and safe zones.

4

Verify: The creative board contains a complete prompt, generation record or wireframe, eleven-row image review, six timed video scenes totalling 15 seconds, voiceover under 38 words, caption, alt text, transcript, rights check, and Status REVIEW. No unverified demonstration result is claimed.

Recap — AI Content Creation and Multimedia Automation

1

You can now: LO3: Produce an aligned multi-platform content and multimedia kit that preserves audience intent, brand voice, factual accuracy, and creative rights

TOPIC 04

Creating and Launching AI-Driven Marketing Campaigns

objectives and segments | KPI trees | message and creative matrices | experiment design | content calendars
| controlled launch

KEY CONCEPTS

Key Concepts — Creating and Launching AI-Driven Marketing Campaigns

1

A campaign objective describes the business outcome; a platform objective is a delivery setting chosen to support it.

2

A KPI tree connects business outcome, conversion event, leading indicator, and diagnostic metric so optimization has a hierarchy.

3

Audience segments should express a need and context, not merely demographic labels or inferred sensitive traits.

4

A message matrix links segment, tension, promise, evidence, objection, and call to action before variations are generated.

5

A useful experiment changes one main variable, defines a control, and states a decision rule before results arrive.

6

A launch workflow validates tracking, creative, links, naming, budgets, approvals, and rollback steps before anything becomes active.

Build a KPI Tree Before Creating Assets

1

Start from the business outcome, then work backward. For a product launch, revenue or purchases may be the outcome. Purchase events are the conversion signal. Landing-page conversion rate and cost per purchase are leading decision metrics. CTR, CPC, reach, and engagement help diagnose why the result moved.

2

This hierarchy prevents the agent from treating every metric as equally important. A high-engagement post may be useful, but it does not automatically prove profitable acquisition. The campaign brief should name the primary decision metric and the guardrails that must not deteriorate.

Visual Model — Build a KPI Tree Before Creating Assets

1

Outcome

Purchases, qualified leads, revenue, or retained value

2

Conversion

The verified event tied to the outcome

3

Decision KPI

CPA, ROAS, conversion rate, or qualified-lead rate

4

Diagnostics

Reach, CTR, CPC, frequency, engagement, and sentiment

5

Guardrails

Spend cap, complaint signal, tracking quality, brand safety

Campaign Message and Creative Matrix

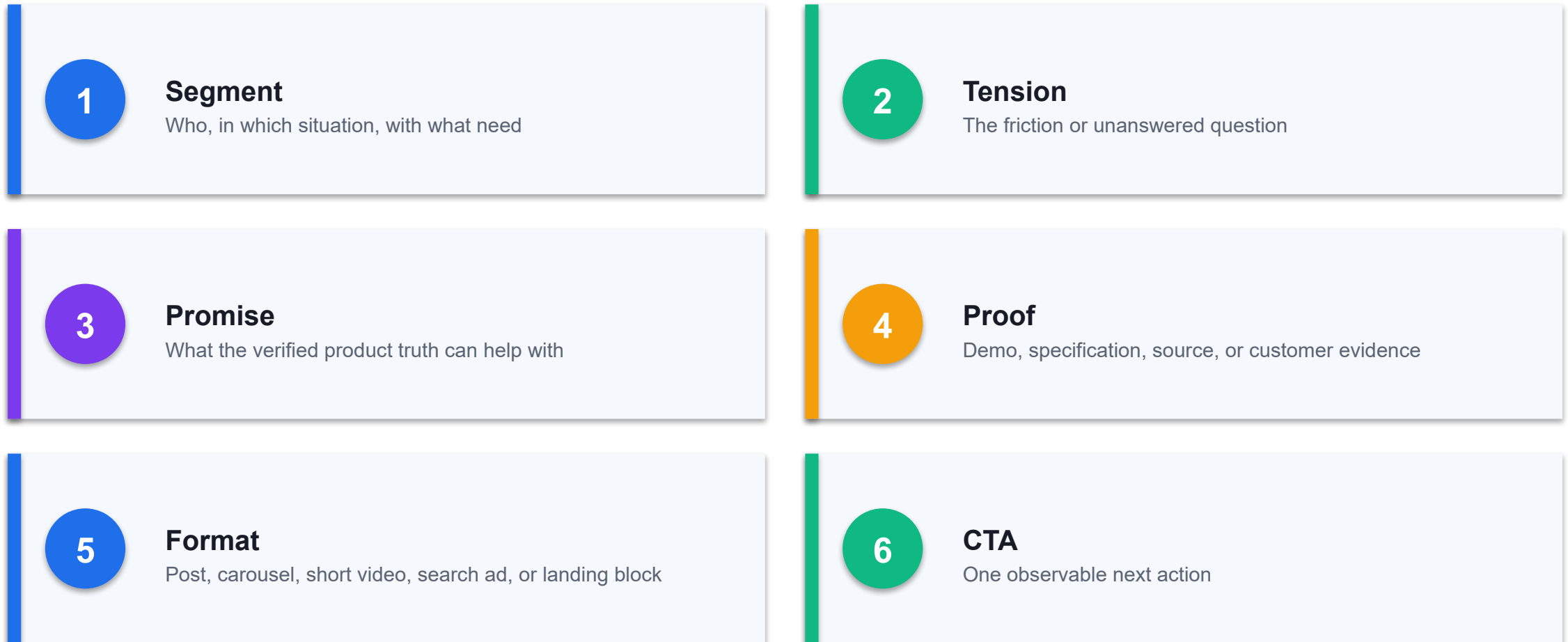
1

A matrix makes variation deliberate. Each row represents a segment and situation; columns hold the audience tension, promise, evidence, objection response, format, and call to action. The agent generates inside those cells rather than improvising a new strategy for every post.

2

The matrix also exposes missing evidence. If a promise has no support, the content should be rewritten as a demonstration, question, or invitation rather than a factual claim. This keeps creativity tied to what the business can honestly show.

Visual Model — Campaign Message and Creative Matrix



The Controlled Launch State Machine

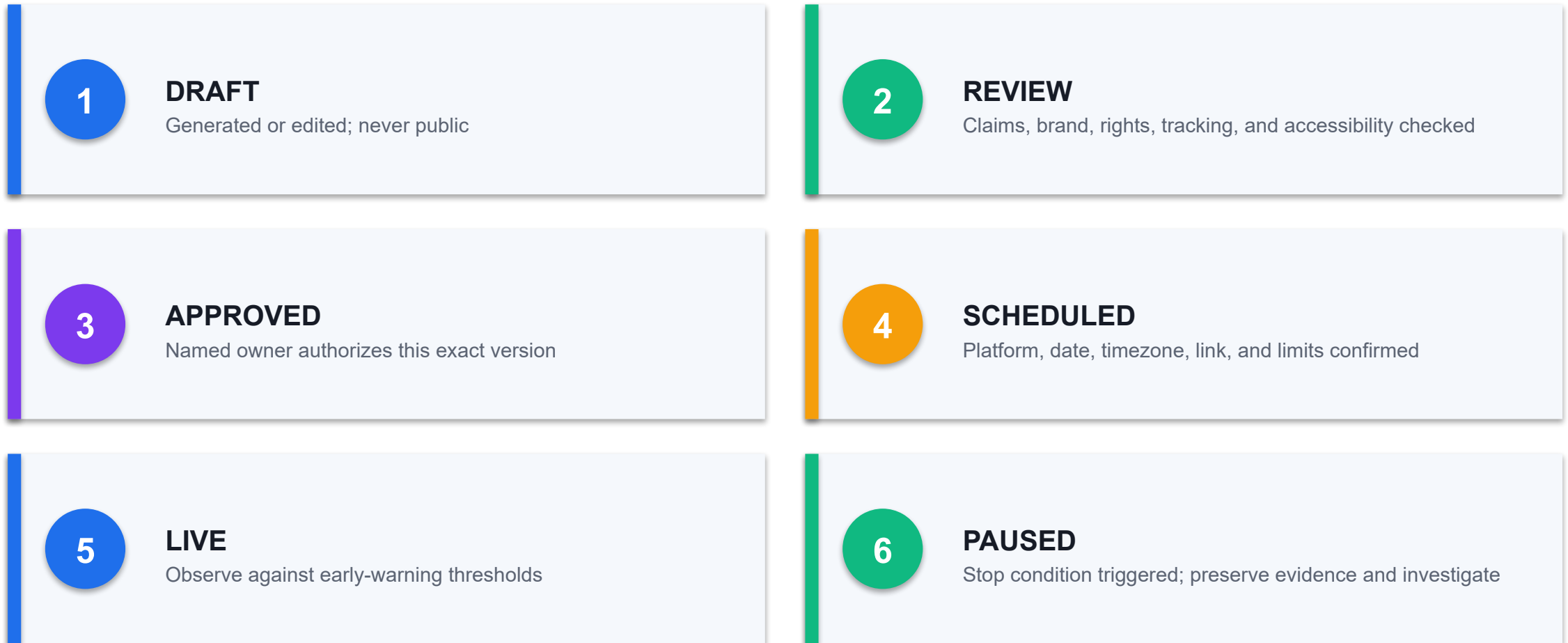
1

A simple state model prevents accidental publication: DRAFT, REVIEW, APPROVED, SCHEDULED, LIVE, PAUSED, and RETIRED. Only named roles may change a state, and the change should capture who, when, why, and which version moved.

2

Before LIVE, verify links, tracking, creative, copy, targeting boundaries, budget limits, date and timezone, and response ownership. A launch plan also defines the first observation window and the condition that will pause delivery.

Visual Model — The Controlled Launch State Machine



Hands-On Labs — Creating and Launching AI-Driven Marketing Campaigns

Lab 7

- Build the Campaign Blueprint, KPI Tree, and Experiment Map

Lab 8

- Create the Content Calendar and Run a Controlled Launch Dry-Run

Build the Campaign Blueprint, KPI Tree, and Experiment Map

LAB 7

You connect the Aurora business objective to segments, messages, creative formats, KPIs, guardrails, and one controlled experiment. The blueprint prevents the agent from optimizing a convenient platform metric that is disconnected from the launch outcome.

You'll build

04-launch/07-campaign-blueprint.md containing an objective tree, segment-message matrix, channel roles, KPI dictionary, experiment card, and ownership map.

Tools: Aurora brief, Labs 5-6 artifacts, document editor, classroom-approved AI assistant

Build Path & Verification — Lab 7

1

1. Connect business outcome, conversion, decision KPIs, diagnostics, and guardrails.

2

2. Align each segment with a verified message, evidence, format, and CTA.

3

3. Predeclare one controlled experiment and its decision rule.

4

Verify: The blueprint links outcome, decision KPIs, diagnostics, and guardrails. EXP-01 changes one hook, requires 10,000 impressions and 150 clicks per arm over seven days, and promotes only on at least 10% CTR lift with intact guardrails.

Create the Content Calendar and Run a Controlled Launch Dry-Run

LAB 8

You schedule eight draft content items across one week, attach owners and evidence, and walk each item through a launch-state dry-run. A formula prevents an incomplete row from entering review, while a zero-LIVE check proves nothing was published during class.

You'll build

04-launch/08-content-calendar.xlsx and 04-launch/08-launch-checklist.md with eight versioned draft items, readiness flags, approval ownership, observation windows, and rollback steps

Tools: Spreadsheet application, Labs 5-7 artifacts, document editor

Build Path & Verification — Lab 8

1. Create eight versioned calendar items with owners, evidence, and rollback versions.

2. Apply a readiness formula and align control/treatment rows.

3. Dry-run one review transition and verify LIVE Count remains zero.

4. Verify: The calendar has eight unique item IDs, complete dates/times/platforms/versions/owners, two aligned experiment arms, and no blank rollback version. CAL-01 is REVIEW with a documented hold reason; all other rows are DRAFT. LIVE Count equals 0 and the checklist covers all eleven sections.

Recap — Creating and Launching AI-Driven Marketing Campaigns

1

You can now: LO4: Design an AI-assisted campaign blueprint, experiment plan, content calendar, and controlled launch workflow tied to objectives and KPIs

TOPIC 05

Agentic AI for Social Media Performance Monitoring

analytics connectors | metric definitions | dashboards | sentiment and themes | pattern detection | evidence-linked feedback

Key Concepts — Agentic AI for Social Media Performance Monitoring

1

A connector should preserve metric name, platform, date window, timezone, object level, and retrieval time so comparisons are interpretable.

2

Reach counts unique exposure while impressions count displays; clicks and engagements need a declared denominator before a rate is compared.

3

CTR, engagement rate, conversion rate, and cost metrics answer different questions and should not be collapsed into one success score without rationale.

4

Sentiment labels are useful only with themes, representative examples, uncertainty, and a human review path for ambiguous or high-risk feedback.

5

Pattern detection compares like with like: platform, audience, format, objective, and time window should be controlled before a winner is named.

6

The feedback loop should pass evidence and a bounded recommendation to content generation, not merely 'make it better.'

The Marketing Metric Ladder

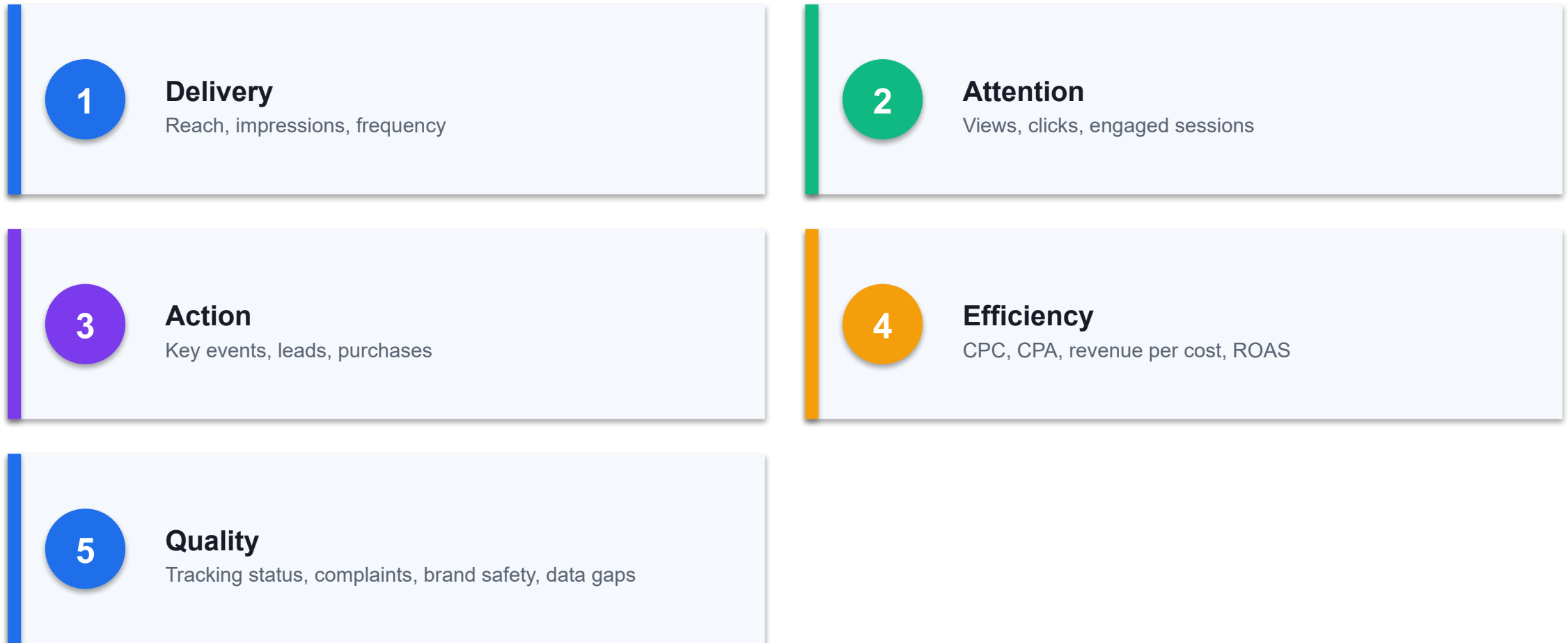
1

Delivery metrics such as reach and impressions show exposure. Attention metrics such as video views and clicks show response. Action metrics such as landing visits and purchases show progression. Efficiency metrics such as CPC, CPA, and ROAS connect those outcomes to cost.

2

Always write the formula and denominator. CTR is clicks divided by impressions. Conversion rate must state whether it uses clicks, sessions, or another base. ROAS is attributed conversion value divided by ad spend and depends on the chosen attribution rules.

Visual Model — The Marketing Metric Ladder



Observation, Interpretation, Recommendation

1

An observation is directly supported by data: Post P10 had the highest clicks in the sample. An interpretation proposes a reason: the launch offer and product image may have increased action. A recommendation proposes a bounded next step: reuse the offer structure in a controlled variation while keeping the audience and placement stable.

2

Agents often overstate interpretation as fact. Requiring these three labelled fields, an evidence reference, and a confidence note makes the reasoning reviewable and easier to challenge.

Visual Model — Observation, Interpretation, Recommendation

1

Observe

State the metric or comment pattern and cite the rows

2

Interpret

Offer a plausible explanation and alternatives

3

Recommend

Specify one reversible next action

4

Verify

Name the KPI, window, and result that would support the idea

Sentiment with Themes and Escalation

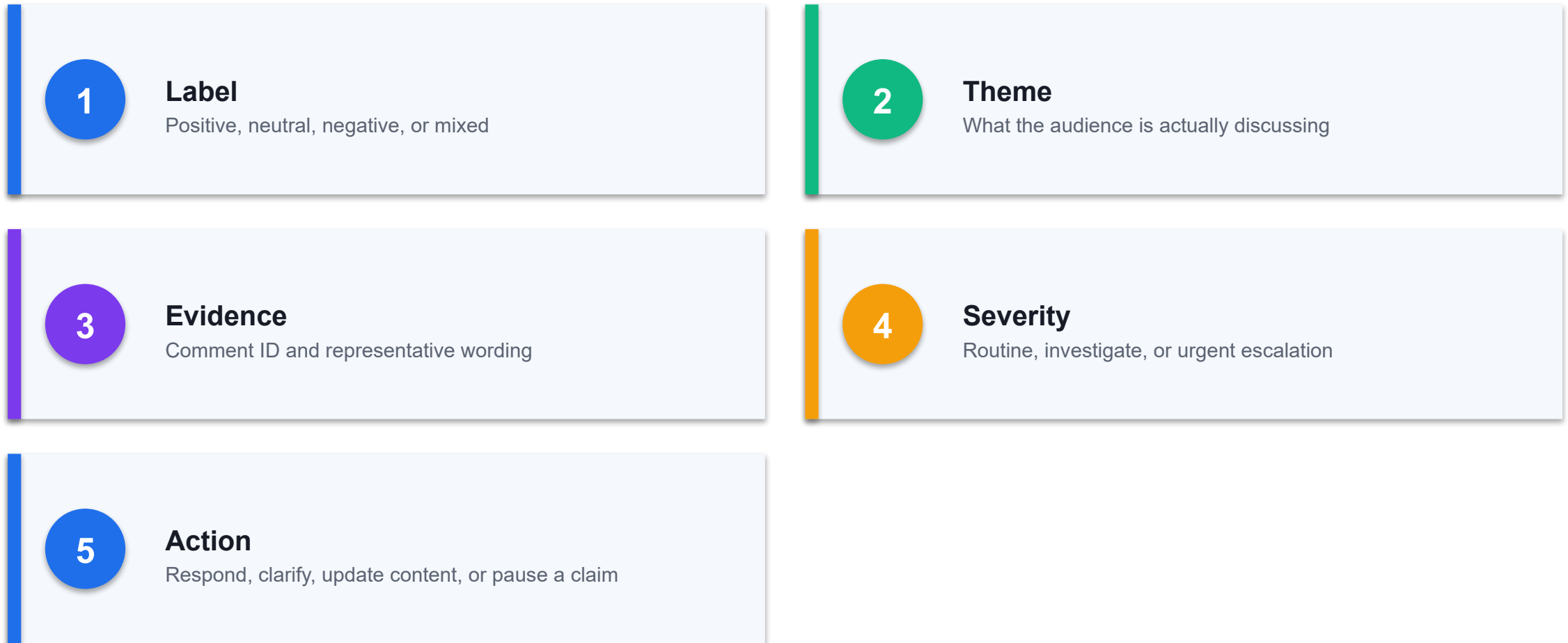
1

Positive, neutral, and negative labels alone are too shallow for action. Add a theme such as price, leak concern, colour request, claim trust, or cleaning question. Preserve a short comment excerpt or ID so a reviewer can see the evidence.

2

Route legal, safety, privacy, discrimination, crisis, or unsupported-claim concerns to a human owner. Do not let an automated reply invent policy or product facts. The useful output is a queue with severity, theme, suggested owner, and a draft response clearly marked for review.

Visual Model — Sentiment with Themes and Escalation



Hands-On Labs — Agentic AI for Social Media Performance Monitoring

Lab 9

- Build a Social Performance Dashboard with Defined Metrics

Lab 10

- Analyse Audience Feedback and Produce an Insight-to-Content Memo

Build a Social Performance Dashboard with Defined Metrics

LAB 9

You calculate platform-neutral performance rates from ten simulated, unpublished posts, validate the denominators, and build a compact dashboard. You separate descriptive findings from causal claims so the dashboard becomes a decision aid rather than a leaderboard.

You'll build

05-monitor/09-social-performance-dashboard.xlsx containing raw data, calculated metrics, a summary table, two charts, metric definitions, and a three-part insight note.

Tools: Spreadsheet application, Aurora post performance CSV, Lab 7 KPI Dictionary

Build Path & Verification — Lab 9

1

1. Validate ten post records and calculate engagements, CTR, and rates.

2

2. Aggregate with weighted formulas and build two labelled charts.

3

3. Write an evidence-linked insight that separates fact from hypothesis.

4

Verify: Ten unique rows calculate correctly; P10 leads clicks and P08 has about 0.68% CTR. The summary uses weighted denominators, charts are labelled, and the insight separates observation from interpretation while marking P08 DO NOT REUSE.

Analyse Audience Feedback and Produce an Insight-to-Content Memo

LAB 10

You classify twelve synthetic comments by sentiment, theme, severity, and owner, then review the AI labels against the original text. You combine comment evidence with the dashboard to propose one bounded content revision and one experiment.

You'll build

05-monitor/10-comment-analysis.xlsx and 05-monitor/10-insight-to-content-memo.md with reviewed labels, escalation routes, representative evidence, and a controlled next action.

Tools: Aurora comments CSV, Lab 9 dashboard, spreadsheet application, classroom-approved AI assistant

Build Path & Verification — Lab 10

1

1. Classify twelve synthetic comments using a strict output schema.

2

2. Review every label and route claim, health, or product uncertainty.

3

3. Combine comment and performance evidence into one bounded next experiment.

4

Verify: All twelve comment IDs have reviewed labels and reviewer notes. C06, C07, and C11 require human review. The memo cites P08, P10, C01, C04, C06, and C07; separates observation from interpretation; marks P08 / LEGACY-FAIL-01 DO NOT REUSE; and proposes EXP-01 without asserting an unperformed result.

Recap — Agentic AI for Social Media Performance Monitoring

1

You can now: LO5: Calculate and interpret social performance metrics, analyse synthetic audience feedback, and turn findings into evidence-linked recommendations

TOPIC 06

Agentic AI for Advertising Optimization and Scaling

paid-media metrics | anomaly detection | diagnosis trees | A/B experiments | budget and targeting controls | autonomous-loop design

Key Concepts — Agentic AI for Advertising Optimization and Scaling

1

An anomaly is a meaningful deviation from an expected range; it is a signal to diagnose, not permission to change a budget.

2

Paid-media diagnosis should check tracking, delivery, audience, creative, landing experience, and conversion lag before choosing a remedy.

3

CPA equals spend divided by conversions; ROAS equals attributed conversion value divided by spend, and both inherit attribution limitations.

4

A/B testing requires a control, one main change, comparable traffic, a predeclared KPI, an observation window, and a decision rule.

5

Scaling should be bounded by spend caps, pacing rules, minimum evidence, reversibility, cooldown periods, and human approval thresholds.

6

A mature optimization loop logs the data snapshot, hypothesis, proposed action, approver, execution result, and rollback state.

Diagnose Before You Optimize

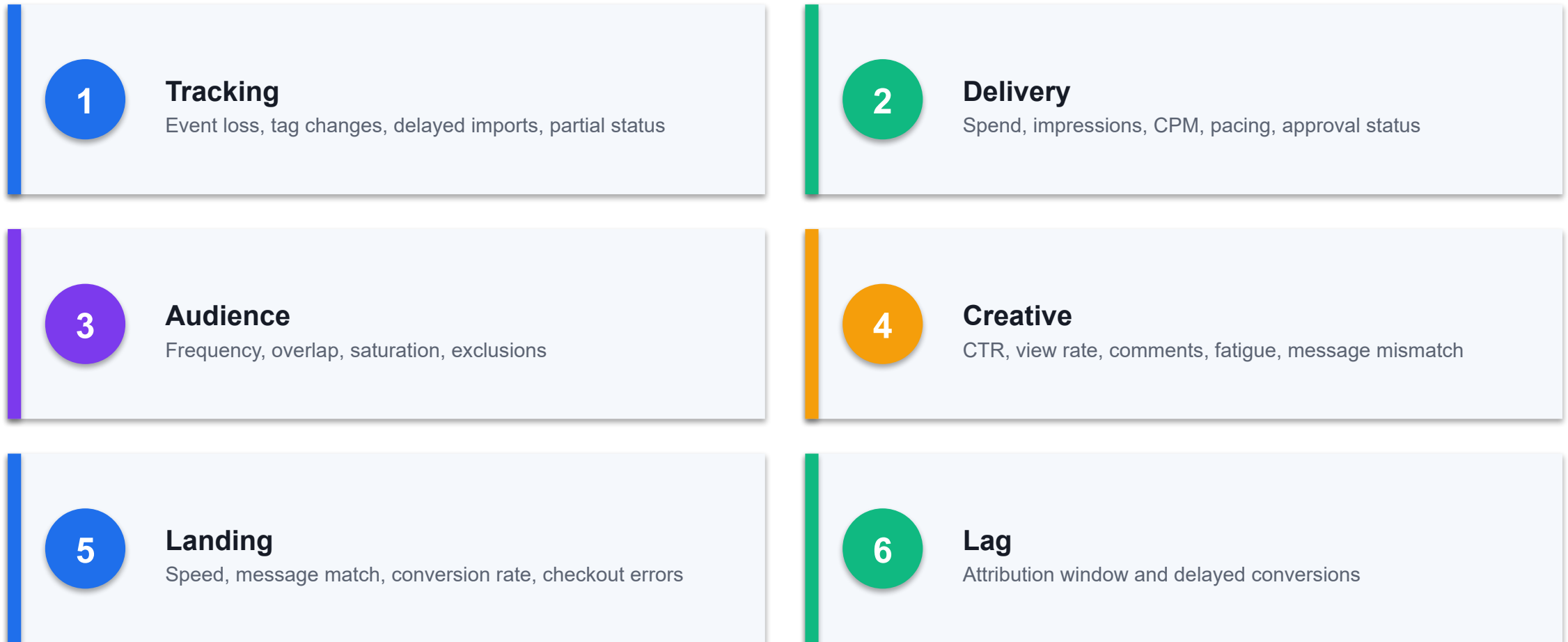
1

Start with tracking quality because every downstream metric depends on it. Then inspect delivery and auction signals, audience saturation, creative response, landing conversion, and conversion lag. Compare the changed segment with a stable reference instead of reacting to the total account number.

2

A good diagnosis memo names the symptom, evidence, competing explanations, missing data, and safest next test. If tracking is partial, the correct recommendation is usually to hold major changes and repair measurement first.

Visual Model — Diagnose Before You Optimize



Design a Decision-Ready Experiment

1

An experiment starts with a causal question, not two random variations. Keep the audience, placement, timing, and offer stable while changing one primary creative or bidding variable. Define the control, treatment, primary KPI, guardrails, minimum runtime, and decision rule before launch.

2

Early differences can be noise. Use the platform's experiment workflow when available, keep control and treatment metrics separate, and avoid declaring a winner from a few conversions. The action after the test may be promote, continue, stop, or redesign.

Visual Model — Design a Decision-Ready Experiment

1

Hypothesis

Because insight X, changing Y should improve KPI Z

2

Control

Current approved version

3

Treatment

One primary change

4

Guardrails

Spend, complaints, tracking, and secondary KPI limits

5

Window

Predeclared runtime and conversion-lag allowance

6

Decision

Promote, continue, stop, or redesign

Bounded Optimization Loop

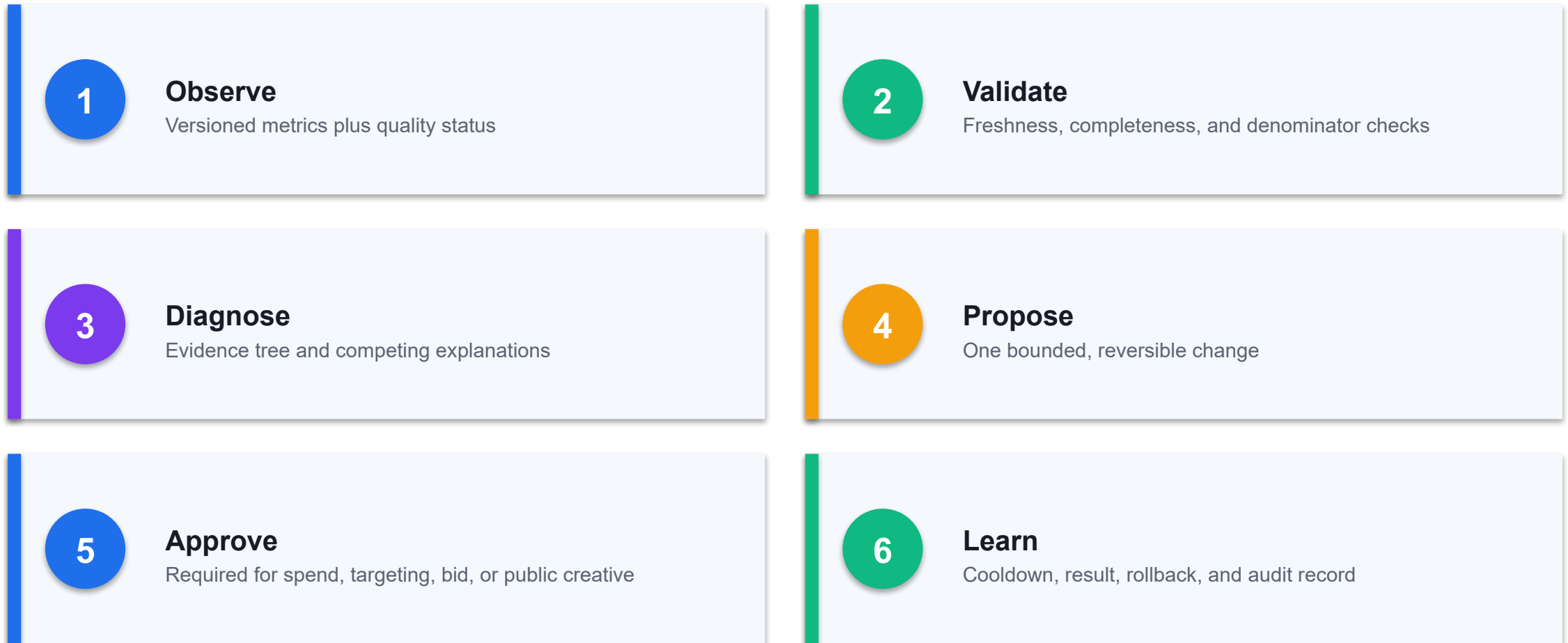
1

The loop collects a versioned snapshot, validates data quality, detects a threshold breach, drafts a diagnosis, proposes one reversible action, and waits for approval when the action affects spend, targeting, or public creative. After execution it observes a cooldown window and compares the result with the rollback condition.

2

Every cycle should be idempotent: rerunning the same input should not apply the same change twice. Store an action ID and current state, enforce daily and weekly caps, and route repeated failures or conflicting evidence to a human owner.

Visual Model — Bounded Optimization Loop



Hands-On Labs — Agentic AI for Advertising Optimization and Scaling

Lab 11

- Diagnose Paid-Media Anomalies Before Recommending a Change

Lab 12

- Design and Simulate a Human-Governed Optimization Loop

Diagnose Paid-Media Anomalies Before Recommending a Change

LAB 11

You calculate CTR, click conversion rate, CPA, and ROAS for eighteen synthetic, unpublished paid-media rows, flag target or tracking breaches, and diagnose the deteriorating commuter ad set. The workflow checks data quality before creative or budget action.

You'll build

06-optimize/11-paid-media-diagnostics.xlsx and 06-optimize/11-diagnosis-memo.md with calculated metrics, anomaly flags, a diagnosis tree, evidence, and a safest-next-test recommendation.

Tools: Spreadsheet application, Aurora ad performance CSV, Aurora targets, Lab 10 memo

Build Path & Verification — Lab 11

1

1. Calculate CTR, conversion rate, CPA, and ROAS for eighteen ad rows.

2

2. Flag target and tracking breaches, then compare the deteriorating segment over time.

3

3. Diagnose tracking first and propose no live change from partial data.

4

Verify: All 18 formulas fill down. The 24-25 July commuter rows show partial tracking; 25 July breaches CPA and ROAS targets. Campaign-day spend stays within SGD 300, and the memo holds budget and targeting changes until tracking is repaired.

Design and Simulate a Human-Governed Optimization Loop

LAB 12

You assemble the course artifacts into a complete observe-validate-diagnose-propose-approve-learn loop. Three tabletop cases test data-quality stops, bounded experiments, approval gates, spend caps, cooldowns, idempotency, audit evidence, and rollback.

You'll build

06-optimize/12-optimization-runbook.md containing the loop, rule table, action schema, three simulation logs, approval matrix, stop conditions, cooldown, and rollback procedure.

Tools: All Aurora campaign checkpoints, document editor, classroom-approved AI assistant

Build Path & Verification — Lab 12

1

1. Define the loop, input contract, rules, permissions, cooldown, and rollback.

2

2. Simulate data-quality hold, experiment proposal, and bounded scaling cases.

3

3. Test idempotency and audit evidence before setting the runbook to v1.0.

4

Verify: Case A holds; Case B proposes one controlled experiment; Case C proposes SGD 297 with approval, cooldown, and rollback. Duplicate action IDs return SKIP DUPLICATE. RULE-SCALE requires two seven-day cycles, at least 30 purchases each, ROAS at least 2.5, clean data, and intact guardrails.

Recap — Agentic AI for Advertising Optimization and Scaling

1

You can now: LO6: Diagnose paid-media performance, propose bounded experiments, and design a human-governed optimization loop with stop conditions and audit evidence



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Agent Architecture

Explain agentic marketing architecture and define goals, tools, memory, approval gates, and operational guardrails for a marketing agent.

2

Trend Intelligence

Build a repeatable trend-discovery workflow that evaluates freshness, evidence, brand fit, and risk before recommending an opportunity.

3

Content Automation

Produce an aligned multi-platform content and multimedia kit that preserves audience intent, brand voice, factual accuracy, and creative rights.

4

Campaign Orchestration

Design an AI-assisted campaign blueprint, experiment plan, content calendar, and controlled launch workflow tied to objectives and KPIs.

5

Performance Learning

Calculate and interpret social performance metrics, analyse synthetic audience feedback, and turn findings into evidence-linked recommendations.

6

Optimization Control

Diagnose paid-media performance, propose bounded experiments, and design a human-governed optimization loop with stop conditions and audit evidence.

Put the System into Practice

1

Replace the fictional brief with an approved low-risk campaign brief from your organisation, keeping the same data fields and gates.

2

Pilot only the read, calculate, and draft stages before connecting any publishing or advertising tool.

3

Create a metric dictionary with owners, formulas, platform fields, windows, and known limitations.

4

Run one controlled content experiment and document the predeclared decision rule before reviewing results.

5

Review permissions, prompts, connectors, logs, and stop conditions at a regular operating cadence.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Your course material stays available at <https://lms-tms.tertiaryinfotech.com/>.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

C1798 COMPLETE

Thank You!

You can now design agentic marketing pipelines that move quickly while keeping evidence, ownership, and control visible.